

Name: _____

In-class Quiz #03: Condition Codes

Condition Codes

Please predict the values of the listed condition flags after running the following instructions in each problem, which are assumed to run on a **16-bit** microprocessor in Thumb mode if not specified. If you think a flag cannot be determined, denote "X". **Please show your work.**

10pts each.

1. MOV16 R1, #0xABAB
 MOV16 R2, #0x1212
 ADDS R0, R1, R2

N: _____; Z: _____; C: _____; V: _____.

2. MOV16 R1, #0x5A00
 MOV16 R2, #0x7E00
 ADDS R0, R1, R2

N: _____; Z: _____; C: _____; V: _____.

3. MOV16 R1, #0x1100
 MOV16 R2, #0xFF00
 ADDS R0, R1, R2

N: _____; Z: _____; C: _____; V: _____.

4. MOV16 R1, #0xFF00
 MOV16 R2, #0xDD00
 ADDS R0, R1, R2

N: _____; Z: _____; C: _____; V: _____.

5. MOV16 R1, #0x7700
 MOV16 R2, #-0xA000
 ADDS R0, R1, R2

N: _____; Z: _____; C: _____; V: _____.

6. MOV16 R1, #0xADAD
 MOV16 R2, #0x1212
 SUBS R0, R1, R2

N: _____; Z: _____; C: _____; V: _____.

7. MOV16 R1, #0x1000
 MOV16 R2, #0xF0F0
 SUBS R0, R1, R2

N: _____; Z: _____; C: _____; V: _____.

8. MOV16 R1, #0x00F0
 MOV16 R2, #0x70F0
 SUBS R0, R1, R2

N: _____; Z: _____; C: _____; V: _____.

9. (32-bits) Fill in the blank to set the flag N (N=1) and clear the flag Z (Z=0). Show your work.

```
MOV32    R1, #0x40000000
MOV32    R2, _____
MULS     R2, __, __
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10. (32-bits) Fill in the blank to set the flag C (C=1) and Z (Z=1) and clear the flag N (N=0) and (V=0). Show your work.

```
MOV32    R1, _____
MOV32    R2, _____
ADDS     R0, __, __
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11. (Extra 5pts) (32-bits) Fill in the blank to set the flag C (C=1), Z (Z=1) and V (V=1), and clear the flag N (N=0). Show your work.

```
MOV32    R1, _____
MOV32    R2, _____
_____  R0, __, __
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